

FRANKLIN, VA, USA

WMO: 723083

Lat: 36.698N

Lon: 76.903W

Elev: 41

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 13763

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.6	22.8	2.7	6.3	24.5	8.5	8.5	29.1	21.0	56.6	18.9	52.7	3.1	350	0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	96.8	77.8	93.1	76.9	90.9	76.1	80.6	90.9	79.3	89.3	78.2	87.6	6.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.6	148.9	85.1	76.8	139.6	83.5	75.2	132.5	82.5	43.9	92.7	42.8	89.5	41.6	88.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.8	15.5	12.8	DB	10.0	99.3	7.8	4.2	4.4	102.3	-0.2	104.7	-4.6	107.0	-10.2	110.0	(4)
(5)			WB	9.1	82.7	7.2	1.7	3.9	83.9	-0.2	84.9	-4.3	85.8	-9.5	87.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.8	41.0	43.8	50.2	60.1	68.4	76.3	79.9	78.4	72.8	61.8	51.1	44.7	(6)	
	DBStd	15.98	11.06	9.82	9.90	8.88	7.50	5.99	4.77	4.66	5.86	8.23	8.82	9.37	(7)	
	HDD50	1003	323	221	123	14	0	0	0	0	0	7	92	223	(8)	
	HDD65	3299	745	595	468	201	54	3	0	0	10	163	426	634	(9)	
	CDD50	4944	44	48	130	316	572	790	928	880	683	372	123	58	(10)	
	CDD65	1764	2	2	10	53	161	343	463	415	243	62	7	3	(11)	
	CDH74	17240	4	14	113	554	1568	3498	4890	4050	1974	511	50	14	(12)	
	CDH80	7282	0	0	19	155	559	1572	2338	1806	701	129	3	0	(13)	
(14) Wind	WSAvg	5.1	5.6	5.9	6.4	6.6	5.5	4.9	4.4	3.9	4.2	4.2	4.6	4.8	(14)	
(15) Precipitation	PrecAvg	48.2	3.6	3.0	4.0	3.1	4.0	4.1	5.5	4.9	5.4	3.7	3.3	3.4	(15)	
	PrecMax	63.1	6.4	6.4	9.4	6.8	6.9	10.0	11.1	13.1	19.1	7.8	8.5	7.2	(16)	
	PrecMin	34.5	0.9	0.9	0.5	0.8	1.3	1.6	1.5	1.4	0.6	0.0	0.7	1.3	(17)	
	PrecStd	7.7	1.5	1.4	1.9	1.5	1.5	2.2	2.4	2.8	4.1	2.0	2.1	1.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.0	75.4	82.1	89.7	93.0	99.1	100.3	98.8	93.0	88.5	79.4	75.2	(19)	
		MCWB	63.5	63.4	65.4	68.7	73.8	77.4	78.4	79.8	75.3	72.9	66.5	64.5	(20)	
	2%	DB	68.9	70.5	77.0	84.1	89.8	94.4	96.9	94.6	90.1	83.3	74.6	70.2	(21)	
		MCWB	62.2	59.7	62.2	66.9	72.6	76.3	78.5	77.9	75.2	71.3	64.3	64.5	(22)	
	5%	DB	63.8	65.8	72.5	81.1	86.1	91.1	93.1	91.3	87.7	80.0	71.6	65.5	(23)	
		MCWB	58.4	57.7	59.4	65.6	71.4	75.7	77.8	76.9	74.3	68.6	63.5	61.3	(24)	
	10%	DB	59.2	61.5	67.9	76.0	82.3	89.6	90.8	89.8	84.0	75.4	66.2	61.4	(25)	
		MCWB	53.3	53.1	58.2	63.2	69.4	75.1	76.9	76.6	72.7	66.9	60.1	56.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	66.1	68.1	72.5	77.2	80.6	82.4	82.4	79.8	76.3	70.4	68.9	(27)	
		MCDB	70.6	72.3	77.5	82.0	89.2	92.1	93.2	94.5	84.8	81.9	75.2	72.6	(28)	
	2%	WB	63.3	63.1	65.6	70.0	74.9	78.5	80.8	80.4	77.9	74.0	67.2	65.3	(29)	
		MCDB	66.9	67.2	73.1	80.1	84.5	90.0	91.4	90.7	84.7	79.5	72.1	68.8	(30)	
	5%	WB	59.5	59.6	63.1	67.8	73.1	77.4	79.6	79.0	76.4	71.6	64.4	62.1	(31)	
		MCDB	63.3	65.5	68.7	77.1	81.9	88.3	90.3	88.2	83.2	77.9	69.0	64.9	(32)	
	10%	WB	54.8	54.6	60.0	65.5	71.5	76.3	78.4	77.8	75.0	68.9	61.7	57.6	(33)	
		MCDB	58.4	59.7	67.0	73.4	80.0	86.0	88.3	86.1	81.3	74.3	66.3	61.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	19.4	21.2	22.5	23.6	21.9	20.9	19.9	19.7	20.1	22.5	22.6	20.2	(35)	
		MCDBR	24.6	26.8	27.7	27.1	25.0	23.5	22.5	22.3	23.0	25.1	26.0	23.4	(36)	
	5% WB	MCWBR	19.0	18.7	16.0	13.1	11.4	9.4	8.9	9.0	10.2	13.6	17.1	17.8	(37)	
		MCWBR	22.6	24.9	23.3	23.2	21.2	20.6	20.0	19.4	18.4	19.7	21.0	21.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.319	0.347	0.391	0.437	0.491	0.518	0.511	0.406	0.357	0.322	0.308	(40)	
		taud	2.523	2.480	2.429	2.316	2.218	2.116	2.061	2.065	2.355	2.481	2.529	2.549	(41)	
	Ebn at Noon		284	292	292	282	270	254	246	245	268	274	274	275	(42)	
		Edh at Noon	25	30	34	40	45	50	52	51	36	29	25	23	(43)	
(44) All-Sky Solar Radiation	RadAvg		765	997	1316	1676	1877	2015	1913	1727	1408	1127	855	677	(44)	
	RadStd		62	112	104	128	155	127	94	96	118	137	69	55	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (63 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+302	+200	

Nomenclature: See separate page